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SARDAR PATEL UNIVERSITY
External Examination (CBCS)
M. Sc. - Information Technology (Integrated)
VIth - Semester (CBCS)
PS06CIIT03 - Computer Graphics
18th April, Tuesday - 2017

Time : 02:00 pm to 5:00 pm

Total Marks :70

Q-1 Select an appropriate option.

10

1. _____ is not the application of Computer Graphics.
(a) Computer Aided Design (b) Presentation Graphics
(c) Power Point (d) Computer Art
2. _____ color display method is used in Random scan display.
(a) Shadow mask (b) Beam Penetration
(c) Color display (d) None of the above.
3. _____ is a primary input device for entering string.
(a) Mouse (b) Joystick (c) Trackball (d) Keyboard
4. DDA Stands for _____.
(a) Data Drawing Algorithm (b) Digital Drawing Algorithm
(c) Digital Differential Analyzer (d) Digital Drawing Analyzer
5. _____ is a single character that can be displayed in different colors and in different sizes.
(a) Marker Symbol (b) Font (c) Character (d) Layout
6. _____ Style generated with a line-drawing routine as a closed polyline.
(a) Hatch (b) Solid (c) Patterned (d) Hollow
7. Rotation point is also known as _____.
(a) Rotation angle (b) Rotation axis
(c) Pivot point (d) Translation vector
8. A _____ is a transformation that produces a mirror image of an object.
(a) Translation (b) Reflection (c) Rotation (d) Shear
9. _____ indicates the selected frame number, the current frame rate, and the elapsed time to the current frame.
(a) Layer (b) Timeline (c) Frame (d) Keyframe
10. To transform a gradient or bitmap fill, _____ tool is used in FLASH.
(a) Free transform (b) Ink bottle (c) Paint bucket (d) Fill Transform

Q-2	Answer the following questions. (Attempt any TEN)	20
1.	Define terms: (i) Aspect Ratio (ii) Resolution	
2.	State disadvantages of DVST.	
3.	What is Beam Penetration method?	
4.	List the joins used when two line intersect.	
5.	Differentiate: Bitmap font and Outline font.	
6.	Explain flood fill algorithm (4-connected) with diagram.	
7.	Define : (i) Scaling (ii) Shear	
8.	What is clipping? List out all clipping object.	
9.	State use of Eyedropper and Pan tool.	
10.	Define: (i) Masking (ii) Layer	
11.	State the purpose of frame and keyframe.	
12.	State conditions for clipping a point.	
Q-3	Explain in detail applications of Computer Graphics.	10
OR		
Q-3	Write short note: (i) Shadow Mask Method (ii) Software Standards	10
Q-4		
(a)	Write short note on DDA line drawing algorithm.	5
(b)	Write a short note on Character generation.	5
OR		
Q-4		
(a)	What is inside–outside test? Explain odd-even rule with example.	5
(b)	Explain Area fill attributes.	5
Q-5		
(a)	What is geometric transformation? List all 2-D geometric transformation and explain any one in detail.	5
(b)	Write a short note on Cohen-Sutherland line clipping algorithm.	5
OR		
Q-5		
(a)	Explain window-to-viewport coordinate transformation.	5
(b)	Explain in brief Sutherland Hodgeman polygon clipping algorithm.	5
Q-6		
(a)	What is onion skinning? Explain the use of onion skinning in detail.	5
(b)	Explain movie clip mask animation with example.	5
OR		
Q-6		
(a)	Explain frame-by-frame animation with example.	5
(b)	List types of symbols used in FLASH. Explain any one of them in detail.	5

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